

Isolation of Syncytial Knots from Placental Tissue

Final Report

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EXECUTIVE SUMMARY

This project addresses the critical need for an automated, high throughput method to isolate syncytial knots from placental tissue for the Hisey Lab's research into maternal health and complications like preeclampsia. The current method of isolation relies on manual micromanipulation to pick individual knots under a microscope, a process that is labor intensive and low throughput, often requiring up to 24 hours to isolate a sufficient sample. Alternative methods like centrifugation often fail to achieve the high purity and gentle handling required for downstream analysis.

To solve this, we designed and fabricated a Deterministic Lateral Displacement (DLD) microfluidic device. This system uses a precisely calculated array of micropillars to “bump” target particles based on size, effectively separating syncytial knots (50-150 μm) from smaller contaminants. Our design process integrated computational modeling through COMSOL Multiphysics. Based on our results, different designs of DLD chips were created with varying gap sizes and number of rows to find the optimal arrangement of parameters for separation.

The device was tested using polystyrene microspheres as a model system to simulate the size and behavior of the syncytial knots and contaminant particles. A design of experiments (DOE) was developed to evaluate the effects of concentration, flow rate, and device geometry on separation efficiency. However, these tests were not fully completed. During testing, several challenges were identified including air bubble formation within the microfluidic channels, clogging, and inconsistent microsphere behavior at the outlet channels. Adjustments to the experimental approach, specifically reducing the number of inlet channels and manually prefilling the device

using a syringe, helped mitigate these issues and enabled initial proof of concept demonstration that the DLD microfluidic can drive bumping of particles in the size range of syncytial knots.

While these results support the feasibility of the design, quantitative performance metrics such as separation efficiency, purity, and throughput have not yet been established. Future work will therefore focus on completing and refining the DOE to systematically evaluate key parameters as well as quantify device performance using particle counts to determine recovery and sorting accuracy. Additionally, device redesign will target the observed failure modes including modifications to the outlet geometry, incorporation of features to prevent particle deviation from the intended streamline, and integration of a bubble trap to improve flow stability. These steps are necessary to verify that the device is operating under optimal and reproducible conditions.

Following optimization and quantitative verification, the final stage of the project will involve testing with real placental explant samples. At that stage, operating parameters such as flow rate and sample concentration will be adjusted to account for differences between the microsphere model system and biological samples. Overall, this work establishes a promising proof of concept framework for automating syncytial knot isolation with the potential to improve throughput compared to manual methods with further optimization and validation.

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1. INTRODUCTION & PROBLEM DEFINITION

1.1 Clinical and Research Motivation

During pregnancy, the placenta mediates maternal-fetal communication by exchanging gases and nutrients and producing hormones critical for pregnancy maintenance^{1,2}. Cytotrophoblasts (CTBs) within the chorionic villi give rise to extravillous trophoblasts (EVTs), which remodel the syncytiotrophoblast (STB), the primary maternal-fetal exchange layer³. Since the STB cannot divide, it is renewed by fusion of new CTBs, and older nuclei are shed into the maternal circulation as syncytial knots^{4,5}. These knots are clinically significant because their abundance correlates with pregnancy complications such as preeclampsia and fetal growth restriction, making them potential early biomarkers⁶. Syncytial knots contain intact fetal nuclei⁷ and are large and stable relative to other blood components⁸ allowing them to offer opportunities for non-invasive fetal genetic testing and the development of novel isolation platforms. Together, these features highlight the clinical need and biological relevance that motivate methods to isolate syncytial knots for diagnostics and research.

1.2 Syncytial Knots

1.2.1 Syncytial Knot Formation

Syncytial knots (Figure 1) are aggregates of aged nuclei within the STB, representing senescent cells that have permanently exited the cell cycle⁹.

1.Placenta. Cleveland Clinic. [accessed 2025 Oct 12]. <https://my.clevelandclinic.org/health/body/22337-placenta>

2.Placental Hormone - an overview | ScienceDirect Topics. [accessed 2025 Oct 12].

<https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/placental-hormone>

3.Chang C-W, Wakeland AK, Parast MM. 2018. Trophoblast lineage specification, differentiation, and their regulation by oxygen tension. *J Endocrinol.* 236(1):R43–R56. doi:10.1530/JOE-17-0402

4.Mayhew TM. 2014. Turnover of human villous trophoblast in normal pregnancy: What do we know and what do we need to know? *Placenta.* 35(4):229–240. doi:10.1016/j.placenta.2014.01.011

5.Burton GJ, Jones CJP. 2009. Syncytial Knots, Sprouts, Apoptosis, and Trophoblast Deportation from the Human Placenta. *Taiwanese Journal of Obstetrics and Gynecology.* 48(1):28–37. doi:10.1016/S1028-4559(09)60032-2

6.Tomkiewicz J, Darmochwał-Kolarz DA. 2024. Biomarkers for Early Prediction and Management of Preeclampsia: A Comprehensive Review. *Med Sci Monit.* 30:e944104-1-e944104-11. doi:10.12659/MSM.944104

7.Askelund KJ, Chamley LW. 2011. Trophoblast deportation part I: Review of the evidence demonstrating trophoblast shedding and deportation during human pregnancy. *Placenta.* 32(10):716–723. doi:10.1016/j.placenta.2011.07.081

8.Redman CWG, Staff AC. 2015. Preeclampsia, biomarkers, syncytiotrophoblast stress, and placental capacity. *American Journal of Obstetrics and Gynecology.* 213(4, Supplement):S9.e1-S9.e4. doi:10.1016/j.ajog.2015.08.003

9.Jones CJ, Fox H. 1977. Syncytial knots and intervillous bridges in the human placenta: an ultrastructural study. *J Anat.* 124(Pt 2):275–286.



Figure 1. Syncytial Knots in Syncytiotrophoblast⁵

CTBs fuse into the STB, replenishing its nuclei, which then migrate along the villous axis to the syncytial buds. Over several weeks, these nuclei undergo apoptosis where caspases degrade cytoskeletal proteins, DNA is fragmented, and nuclei cluster at the villus tips to form syncytial knots¹⁰. These clusters are eventually extruded into the maternal circulation.

1.2.2 Syncytial Knot Progression During Gestation

Quantitative analysis shows that the proportion of villi containing syncytial knots increases progressively during gestation: 5-9% at 20-25 weeks, 11-15% at 26-33 weeks, ~23% at 36 weeks, and ~28% at term¹¹. A syncytial knot is defined histologically as an aggregate of at least five STB nuclei protruding from the villous surface and not in contact with adjacent villi¹².

1.2.3 Characteristics of Syncytial Knots

Syncytial knots are transcriptionally inactive due to chromatin condensation, yet they reflect the normal turnover of the STB. They are large (50-150 μm) relative to other maternal blood components, maintain

5. Burton GJ, Jones CJP. 2009. Syncytial Knots, Sprouts, Apoptosis, and Trophoblast Deportation from the Human Placenta. *Taiwanese Journal of Obstetrics and Gynecology*. 48(1):28–37. doi:10.1016/S1028-4559(09)60032-2

10. Austgulen R, Chedwick L, Isaksen CV, Vatten L, Craven C. 2002. Trophoblast Apoptosis in Human Placenta at Term as Detected by Expression of a Cytokeratin 18 Degradation Product of Caspase. *Arch Pathol Lab Med*. 126(12):1480–1486. doi:10.5858/2002-126-1480-TAIHPA

11. Loukeris K, Sela R, Baergen RN. 2010. Syncytial knots as a reflection of placental maturity: reference values for 20 to 40 weeks' gestational age. *Pediatr Dev Pathol*. 13(4):305–309. doi:10.2350/09-08-0692-OA.1

12. Fogarty NME, Ferguson-Smith AC, Burton GJ. 2013. Syncytial Knots (Tenney-Parker Changes) in the Human Placenta: Evidence of Loss of Transcriptional Activity and Oxidative Damage. *The American Journal of Pathology*. 183(1):144–152. doi:10.1016/j.ajpath.2013.03.016

structural integrity during extrusion, and serve as a mechanism to isolate senescent nuclei while allowing the STB to continue its metabolic and endocrine functions^{9,10,12}. These properties make syncytial knots distinct and identifiable within the placenta and maternal circulation.

1.3 Engineering Challenge

Although syncytial knots are large and distinct, they are difficult to isolate due to their fragility, rarity, and size overlap with other maternal blood components including single cells, cell clusters, and ECM aggregates around 10-30 μm ⁸. Conventional separation methods including centrifugation, filtration, and immuno-based approaches are ineffective as they either damage the knots or fail to achieve sufficient purity. Developing an effective isolation platform must address both biological constraints and engineering limitations:

- Preserving structural integrity: Syncytial knots must remain intact during handling and processing
- Flow control and mechanical stress: Components of the system must minimize shear forces that could disrupt knot membranes.
- Reproducibility and throughput: The system must consistently isolate sufficient numbers of knots for experimental and diagnostic use.

1.4 Project Objectives

The objective of this project is to develop a method for isolating syncytial knots from placental explant tissue while minimizing contamination from other cellular material. By achieving reliable isolation, this work will advance maternal and fetal health research as syncytial knots serve as indicators of placental function

8.Redman CWG, Staff AC. 2015. Preeclampsia, biomarkers, syncytiotrophoblast stress, and placental capacity. American Journal of Obstetrics and Gynecology. 213(4, Supplement):S9.e1-S9.e4. doi:10.1016/j.ajog.2015.08.003

9.Jones CJ, Fox H. 1977. Syncytial knots and intervillous bridges in the human placenta: an ultrastructural study. J Anat. 124(Pt 2):275–286.

10.Austgulen R, Chedwick L, Isaksen CV, Vatten L, Craven C. 2002. Trophoblast Apoptosis in Human Placenta at Term as Detected by Expression of a Cytokeratin 18 Degradation Product of Caspase. Arch Pathol Lab Med. 126(12):1480–1486. doi:10.5858/2002-126-1480-TAIHPA

12.Fogarty NME, Ferguson-Smith AC, Burton GJ. 2013. Syncytial Knots (Tenney-Parker Changes) in the Human Placenta: Evidence of Loss of Transcriptional Activity and Oxidative Damage. The American Journal of Pathology. 183(1):144–152. doi:10.1016/j.ajpath.2013.03.016

and maternal health. Successfully isolating these structures will enable further study of their role in pregnancy disorders, support non-invasive fetal DNA sequencing, and contribute to the development of rapid diagnostic approaches. Ultimately, refining isolation strategies for syncytial knots has the potential to improve prenatal care, enhance understanding of placental biology, and generate tools for safer, earlier, and more informative maternal health diagnostics.

2. USERS & USE ENVIRONMENT

2.1 The Hisey Lab

The primary user group for this device is the Hisey Lab, a research team specializing in microdevices used for the isolation and study of biological vehicles including various types of extracellular vesicles and interested in placental biology at Northwestern University. Their work focuses on utilizing microfluidics to isolate specific molecular targets including syncytial knots which play an important role in studying the mechanisms of placental function. The lab requires tools that can integrate into their workflow which involves the culturing of placental tissue, isolating specific cellular structures, and performing microscopy based analyses. Their use environment includes biosafety cabinets, imaging suits, and standard laboratory facilities, so the device must be compatible with routine safe handling protocols.

2.2 Research & Clinical Community

Beyond the Hisey Lab, this technology is relevant to the broader placental biology and reproductive health research community. Potential users include academic and clinical scientists studying placental structure, pregnancy complications, and fetal development. Clinicians could also benefit from the technology by using isolated syncytial knots to investigate disorders such as preeclampsia, fetal growth restriction, or maternal vascular malperfusion, enabling research and diagnostic applications that extend beyond the lab.

3. DESIGN REQUIREMENTS

The design requirements for the syncytial knot isolation platform are centered on four key goals outlined in Table 1: maintaining membrane integrity, achieving high purity, increasing throughput, and automating the process. Together, these requirements ensure that isolated syncytial knots remain biologically intact, can be reliably distinguished from contaminants, are obtained efficiently, and reduce labor intensive handling, making the method practical for research and potential diagnostic applications.

Table 1. Summary of requirements and specifications

Requirement	Maintain an intact membrane of the syncytial knots	Obtain an isolated sample of syncytial knots from placental tissue	Increase the throughput	Increase automation of the isolation process and decrease man time
Specification	< 10% cytotoxicity	> 90% purity of syncytial knots	Obtain ~240 knots in < 4 hours	< 1 hour man time per 400mg explant sample, 75% automation
Method of assessment	Lactate dehydrogenase release assay	Fluorescence imaging with a nuclear stain and syncytial markers	Measure the rate of processing in mL/hr	Measure the amount of time required to manually process and set up the isolation

3.1 Functional Requirements

3.1.1 Purity

High purity minimizes contamination from cell clusters and extracellular material which can be similar in size to syncytial knots (~10-30 μm). This allows accurate biological measurements and ensures that observed results reflect the syncytial knots themselves and not contaminants on the micron scale.

3.1.2 Throughput

Current manual isolation is slow, taking 12-24 hours per 240 knots.

Increasing throughput to under 4 hours with the same amount of placental

material will accelerate experiments, allow larger scale studies, and improve efficiency.

3.1.3 Membrane Integrity

Preserving membrane integrity protects the internal molecules, organelles, and genetic material of syncytial knots. Minimizing physical stress like shear forces and chemical stress that could perforate the membrane ensures the sample remains biologically useful for downstream analyses.

3.1.4 Automation

Automating the process reduces labor, limits human error, and increases reproducibility. A self operating system allows researchers to focus on downstream research applications of syncytial knots rather than manual isolation, improving efficiency.

3.2 Quantitative Metrics

3.2.1 Purity

Purity will be assessed through fluorescence imaging using nuclear stains and syncytiotrophoblast specific markers (cytokeratin 7, β -hCG)^{13,14}. The proportion of syncytial knots relative to contaminants in the collected fraction will determine the sample's relative purity aiming for >90% syncytial knots.

3.2.2 Throughput

Throughput will be quantified by the volume of tissue processed per unit time (mL/hr) based on the use of filtered conditioned media from Netwell

13. Pantham P, Chamley LW. 2018. Harvesting and Characterization of Syncytial Nuclear Aggregates Following Culture of First Trimester Human Placental Explants. *Methods Mol Biol.* 1710:155–163. doi:10.1007/978-1-4939-7498-6_12

14. Morosin SK, Delforce SJ, Corbisier de Meaultsart C, Lumbers ER, Pringle KG. 2021. *FURIN and placental syncytialisation: a cautionary tale.* *Cell Death Dis.* 12(7):635. doi:10.1038/s41419-021-03898-z

prepared explant tissue, targeting ~240 knots in under 4 hours, a fivefold increase over current methods.

3.2.3 Membrane Integrity

The integrity of syncytial knot membranes will be measured via a lactate dehydrogenase (LDH) release assay^{15,16}. LDH release indicates cellular damage, allowing quantification of cytotoxicity and verification that the isolation technique preserves knot integrity since perforated knots will release LDH. The target is <10% cytotoxicity from the isolated knots.

3.2.4 Automation

Automation will be measured by researcher labor per 400mg explant including the setup and processing of the tissue. The aim is to reduce manual time to <1 hour (~25% of current method) while maintaining yield and quality.

15.Riss T, Niles A, Moravec R, Karassina N, Vidugiriene J. 2019. Cytotoxicity Assays: In Vitro Methods to Measure Dead Cells. In: Assay Guidance Manual [Internet] [Internet]. [place unknown]: Eli Lilly & Company and the National Center for Advancing Translational Sciences; [accessed 2025 Oct 11]. <https://www.ncbi.nlm.nih.gov/sites/books/NBK540958/>

16.Riss T, Niles A, Moravec R, Karassina N, Vidugiriene J. 2019. Cytotoxicity Assays: In Vitro Methods to Measure Dead Cells. In: Assay Guidance Manual: Eli Lilly & Company and the National Center for Advancing Translational Sciences; [accessed 2025 Oct 11]. <https://www.ncbi.nlm.nih.gov/sites/books/NBK540958/>

4. EVALUATION OF EXISTING TECHNOLOGY

Understanding and improving the isolation of syncytial knots is essential for advancing placental research. Existing techniques vary in specificity, throughput, and ability to preserve knot integrity. While current methods are effective in controlled contexts, limitations in scalability, speed, and reproducibility highlight the need for more efficient, automated, and gentle isolation strategies that can maintain the structural and functional integrity of syncytial knots.

4.1 Manual Isolation

Manual isolation relies on micromanipulation under a microscope using pneumatic injectors. Researchers aspirate individual syncytial nuclear aggregates with a fine glass capillary, controlling liquid flow to avoid structural damage¹⁷.

While highly specific and capable of yielding pure knots, the process is extremely slow, requiring 12–24 hours to isolate ~240 knots from a 12-well plate. This time intensive approach limits experimental throughput and reproducibility, creating a clear need for more streamlined techniques.

4.2 Centrifugation-Based Methods

4.2.1 Simple and Differential Centrifugation

Standard centrifugation separates particles based on size and density, allowing removal of large debris¹⁸. Differential centrifugation improves precision by sequentially pelleting components of increasing density, which can partially separate syncytial knots (50–150 μm) from smaller placental fragments¹⁹. For cell clusters or ECM aggregates, trypsin may potentially help reduce the size of these contaminants.

17.Huppertz B, Kingdom J, Caniggia I, Desoye G, Black S, Korr H, Kaufmann P. 2003. Hypoxia Favours Necrotic Versus Apoptotic Shedding of Placental Syncytiotrophoblast into the Maternal Circulation. *Placenta*. 24(2):181–190. doi:10.1053/plac.2002.0903

18.Particle/cell separation on microfluidic platforms based on centrifugation effect: a review | Microfluidics and Nanofluidics. [accessed 2025 Dec 12]. <https://link.springer.com/article/10.1007/s10404-017-1933-4>

19.Elagamey E, Narula K, Chakraborty N, Chakraborty S. 2020. Extracellular Matrix Proteome: Isolation of ECM Proteins for Proteomics Studies. In: Gupta KJ, editor. *Nitrogen Metabolism in Plants: Methods and Protocols* [Internet]. New York, NY: Springer; [accessed 2025 Oct 8]; p. 155–172. doi:10.1007/978-1-4939-9790-9_14

4.2.2 Density Gradient Separation

Density gradients, often using sucrose, provide finer resolution by allowing particles to migrate to layers matching their density²⁰. This method can preserve knot integrity under slow spin speeds but requires careful handling and preparation, limiting scalability.

While centrifugation based approaches are accessible, they lack the specificity of manual isolation and often require multiple steps to achieve sufficient purity, making them impractical for high throughput studies.

4.3 Sieve-Based Filtration

Sequential sieves can separate knots by size while reducing mechanical stress²¹. Conditioned media passes through decreasing pore sizes, retaining knot sized material while removing larger debris. Use of cushioning buffers like BSA further protects knot structure. The main concern is the non-spherical morphology of syncytial knots and limitations of sieves on the micron scale.

4.4 Immunoassay Based Isolation

Specific antibodies targeting syncytial knot antigens can be coupled to magnetic beads, enabling selective isolation²². Cleavable linkers allow removal of beads post isolation, improving purity and reducing stress on the knots²³. These methods offer high specificity but may require extensive optimization and expensive reagents.

20. Yang H, Murphy A. 2013. Membrane Preparation, Sucrose Density Gradients and Two-phase Separation Fractionation from Five-day-old *Arabidopsis* seedlings. Bio-protocol [Internet]. [accessed 2025 Oct 12] 3(24). doi:10.21769/BioProtoc.1014

21. Urban MA, Romero IC, Sivaguru M, Punyasena SW. 2018. Nested cell strainers: An alternative method of preparing palynomorphs and charcoal. Review of Palaeobotany and Palynology. 253:101–109. doi:10.1016/j.revpalbo.2018.04.002

22. Immunomagnetic Separation - an overview | ScienceDirect Topics. [accessed 2025 Oct 12]. <https://www.sciencedirect.com/topics/medicine-and-dentistry/immunomagnetic-separation>

23. The Importance of Tissue Dissociation Enzymes in Cell & Tissue Culture - LevitasBio. 2021. [accessed 2025 Oct 12]. <https://levitasbio.com/the-importance-of-tissue-dissociation-enzymes-in-cell-tissue-culture/>

4.5 Limitations of Existing Methods

Each current technique involves trade offs between purity, throughput, and knot integrity. Manual picking provides high purity but is slow and labor intensive. Centrifugation and filtration partially isolate knots but cannot fully eliminate contaminants. Immuno-based methods are precise but costly and require extensive preparation. Microfluidics show promise for rapid, gentle, and high throughput processing, yet current designs are optimized for much smaller particles (30–150 nm) rather than the larger 50–150 μm syncytial knots.

Table 2. Comparison of syncytial knot isolation techniques based on performance attributes

Requirement	Manual Picking	Centrifugation	Differential Centrifugation	Gradient Separation	Sieve Filtration	Immunoassay	Microfluidics
Time efficient		X	X		X		X
Purity				X			X
Gentle/minimal stress	X			X	X	X	X
High Throughput		X	X			X	X
Compatible with Imaging	X					X	X
Ease of implementation		X	X		X		X

The limitations detailed in Table 2 collectively motivate the use of microfluidic separation platforms specifically designed for syncytial knots, combining gentle handling, high purity, and scalable throughput to satisfy our requirements.

5. MICROFLUIDIC DESIGN STRATEGY & SELECTION

5.1 *Why Microfluidics*

Microfluidic separation provides a high throughput, gentle, and controllable method for isolating syncytial knots from placental explant tissue. Unlike manual isolation, which is slow and labor intensive, or centrifugation and filtration based methods, which struggle with purity and structural preservation, microfluidics can process complex samples while maintaining knot integrity. Historically they have been used to isolate small particles such as exosomes in the 30-150 nm range and circulating tumor cells in the 10-30 μm range^{24,25}. Since microfluidic systems allow precise tuning of fluid dynamics to separate particles based on size, scaling them up would make them ideal for isolating the 50–150 μm range of syncytial knots.

5.2 *Candidate Approaches*

To address the limitation of conventional isolation, three microfluidic strategies were evaluated as potential candidates for the automated separation of syncytial knots. Each method utilizes distinct principles to achieve size based isolation with the 50-150 μm range.

5.2.1 *Spiral Microfluidic*

Spiral microfluidics utilize inertial lift and Dean drag forces to focus particles into specific streamlines as shown in Figure 2^{26,27}. Rectangular channels create high aspect ratios that drive focusing of larger particles such as syncytial knots in the primary flow within the inside of the curve

24. Wu Y, Wang Y, Lu Y, Luo X, Huang Y, Xie T, Pilarsky C, Dang Y, Zhang J. 2022. Microfluidic Technology for the Isolation and Analysis of Exosomes. *Micromachines* (Basel). 13(10):1571. doi:10.3390/mi13101571

25. Macaraniag C, Luan Q, Zhou J, Papautsky I. 2022. Microfluidic techniques for isolation, formation, and characterization of circulating tumor cells and clusters. *APL Bioeng.* 6(3):031501. doi:10.1063/5.0093806

26. Sen-Dogan B, Demir MA, Sahin B, Yildirim E, Karayalcin G, Sahin S, Mutlu E, Toral TB, Ozgur E, Zorlu O, Kulah H. 2023. Analytical Validation of a Spiral Microfluidic Chip with Hydrofoil-Shaped Pillars for the Enrichment of Circulating Tumor Cells. *Biosensors* (Basel). 13(10):938. doi:10.3390/bios13100938

27. Kato S, Carlson DW, Shen AQ, Guo Y. 2024. Twisted fiber microfluidics: a cutting-edge approach to 3D spiral devices. *Microsyst Nanoeng.* 10:14. doi:10.1038/s41378-023-00642-9

while smaller debris is carried away by secondary flows concentrated at the outside of the curve. While the advantage of this design is the continuous operation without obstacles which minimizes mechanical stress and reduces the risk of clogging, the underlying fluid dynamics are highly complex making the system more difficult to model, optimize, and troubleshoot for precise size based separation.

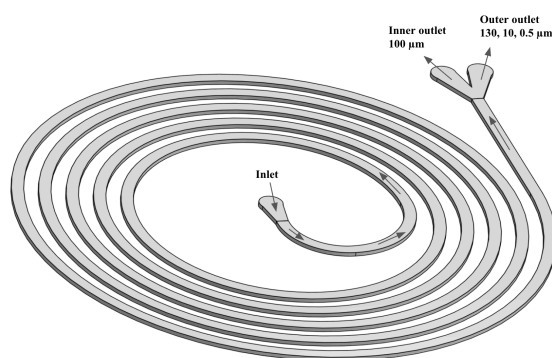


Figure 2. Spiral microfluidic

5.2.2 Cross Flow Microfluidic

Crossflow microfluidics separate particles using a pressure driven flow over micropillar arrays, allowing smaller particles to pass into side channels while larger syncytial knots remain in the primary flow as shown in Figure 3^{28,29}. While this design provides predictable retention of target particles, achieving precise size based discrimination can be challenging as separation depends on flow distribution and pressure gradients rather than strictly defined geometric designs. Other concerns included the

28. Chiu Y-Y, Huang C-K, Lu Y-W. 2016. Enhancement of microfluidic particle separation using cross-flow filters with hydrodynamic focusing. *Biomicrofluidics*. 10(1):011906. doi:10.1063/1.4939944

29. Zhou J, Mukherjee P, Gao H, Luan Q, Papautsky I. 2019. Label-free microfluidic sorting of microparticles. *APL Bioeng*. 3(4):041504. doi:10.1063/1.5120501

complexity of the design with the addition of loops and potential clogging in the side channels.

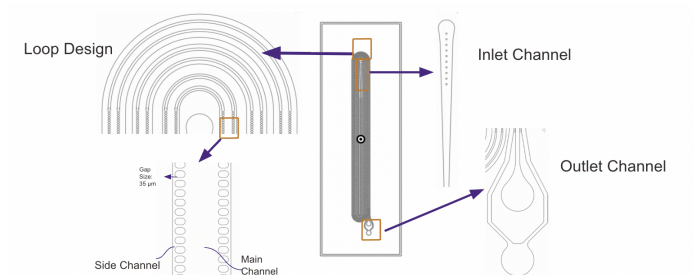


Figure 3. Cross flow microfluidic

5.2.3 Deterministic Lateral Displacement (DLD) Microfluidic

DLD microfluidics separate particles using ordered arrays of posts that laterally displace particles above a critical diameter while allowing smaller particles to move along the bulk flow as shown in Figure 4³⁰. Key parameters such as gap size, post diameter, row shift, and critical diameter can be precisely calculated to optimize separation of the 50–150 μm syncytial knots³⁰. The primary strength of the DLD is its highly predictable geometry driven separation mechanism which enables precise control over particle trajectories. This deterministic behavior makes it well suited for applications requiring size specific discrimination. Preliminary verification using polystyrene microspheres demonstrated displacement of particles within the target size range supporting its feasibility for this application. The main concern is the dense pillar array that could introduce

30.Davis JA. Microfluidic Separation of Blood Components through Deterministic Lateral Displacement.

a risk of clogging if upstream sample preparation does not adequately remove large aggregates or debris.

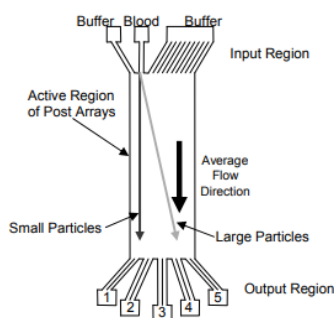


Figure 4. DLD microfluidics ³⁰

5.3 Comparison & Final Selection (DLD)

While spiral and crossflow microfluidics offer potential strategies for size based separation, their complexity and lower control over precise particle trajectories make them less suitable for isolating 50–150 μm syncytial knots. Spiral devices require complex modeling of inertial and Dean flows, and crossflow devices depend on pressure driven separation, which increases the complexity of device design and potential for unexpected variables to affect separation efficiency.

In contrast, DLD provides predictable, geometry driven separation that can be optimized for the target size range. Preliminary COMSOL modeling and microsphere testing support its feasibility and reproducibility. Therefore, DLD was selected as the preferred microfluidic strategy for further development, with future work focusing on optimizing device geometry and minimizing clogging to

³⁰. Davis JA. Microfluidic Separation of Blood Components through Deterministic Lateral Displacement.

enable high purity and high throughput isolation of syncytial knots from placental explant tissue.

6. SYSTEM OVERVIEW

6.1 Overall Workflow

The system is designed as a continuous pipeline that transitions raw biological tissue into a purified sample through a specific sequence of operations as shown in Figure 5. The process begins with tissue extraction, where syncytial knots are shed from cultured placental explants into a liquid media. This media then undergoes preprocessing, involving filtration and enzymatic treatment to clear large aggregates and break down cell clusters. The prepared sample is run through the microfluidic device to sort the knots from smaller contaminants based on their size. The process concludes with a collection phase, where the target knots are retrieved from the device outlet and concentrated for final research applications.

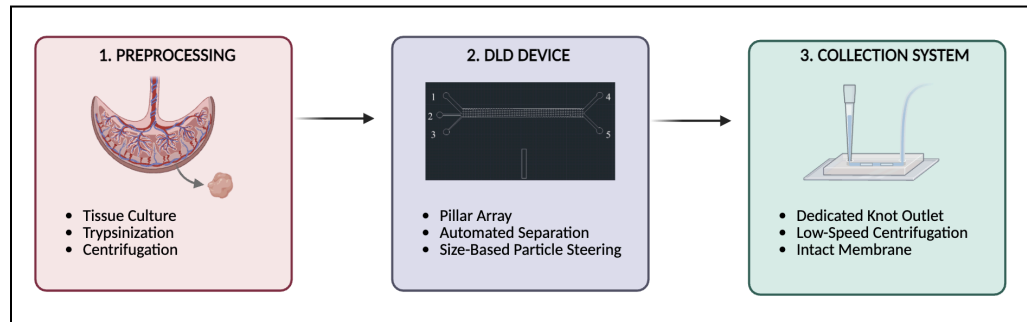


Figure 5. Workflow for isolating and collecting syncytial knots from placental explant tissue

6.2 Key Components

6.2.1 Preprocessing

To ensure consistent flow and prevent device failure, the placental media undergo a multi-step preparation phase. We utilize 400 μm sequential mesh sieves to catch large tissue fragments that would otherwise cause immediate clogging of the microfluidic inlets. Additionally, we plan to

utilize trypsin, an enzymatic treatment, to break down extracellular matrix aggregates³¹. This step ensures that the particles entering the chip are individually suspended and fall within the intended size range ($<150\text{ }\mu\text{m}$).

6.2.2 DLD Device

The core of the system is the DLD microfluidic chip. The device utilizes a physical pillar array to create a separation mechanism that moves particles of calculated sizes along different paths depending on their diameter.

Under the assumption that syncytial knots are the largest structures in the filtered and processed media, the device effectively isolates them from the smaller cellular debris.

6.2.3 Collection System

After separation, the syncytial knots are collected into a dedicated outlet at the end of the DLD chip. Since the knots exit the chip in a diluted buffer, we plan to collect the flow in microcentrifuge tubes and perform low speed centrifugation to pellet the knots. This pellet formation at the bottom of the tube enables easy removal of the supernatant to collect a concentrated sample of knots while ensuring the membranes remain fully intact for analysis.

31. The Importance of Tissue Dissociation Enzymes in Cell & Tissue Culture - LevitasBio |. 2021. [accessed 2026 Mar 19]. <https://levitasbio.com/the-importance-of-tissue-dissociation-enzymes-in-cell-tissue-culture/>

7. DLD DEVICE DESIGN

7.1 Principle of Operation

The DLD microfluidic device separates particles based on size using a structured array of posts as shown in Figure 6. As the sample flows through the device alongside buffer streams, larger particles, such as syncytial knots, are displaced laterally by the post array, following a “bump mode” path, while smaller particles move undisturbed in a zigzag trajectory toward the outlet. The critical diameter (D_C) defines the threshold particle size that determines whether a particle is displaced or follows the lateral flow, enabling precise size based separation of the target particles from smaller debris.

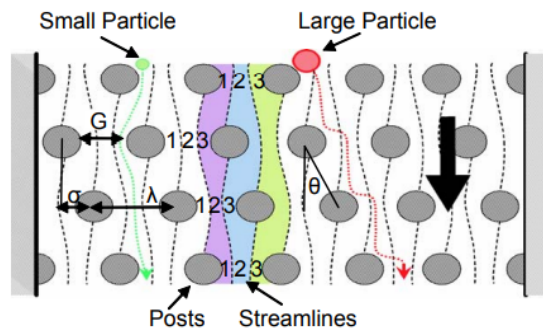


Figure 6. Important parameters in DLD microfluidic design and particle streamlines³⁰

7.2 Critical Parameters

Key parameters defining the DLD microfluidic include:

- **Gap size (G):** Distance between adjacent posts, set to accommodate the full range of target particle sizes.
- **Post diameter (P):** Set to approximately match the gap size for uniform displacement.

30.Davis JA. Microfluidic Separation of Blood Components through Deterministic Lateral Displacement.

- **Row-to-row shift (σ) and shift angle (θ):** Control lateral displacement of larger particles across the array.
- **Relative shift fraction (ϵ):** Determines the number of streamlines and the period of the array required for complete separation.
- **Pillar pitch (λ):** Distance between the centers of consecutive posts.
- **Critical diameter (D_c):** The threshold diameter in which larger particles experience bumping.

Particles smaller than D_c follow a zigzag flow, while those equal to or larger than D_c are displaced into the bump mode, creating distinct separation paths.

7.3 Design Calculations

Design calculations were performed to determine the parameters for a critical diameter of 45 μm which separates the contaminants from the syncytial knots (50–150 μm). The relative shift fraction (ϵ), row shift angle (θ), and row displacement (σ) were calculated for a gap and post diameter of 200 μm .

$$\epsilon = \left(\frac{D_c}{1.4G} \right)^{2.083} \quad (\text{eq. 1})$$

$$\theta = \tan^{-1}(\epsilon) \quad (\text{eq. 2})$$

$$\sigma = \epsilon * \lambda \quad (\text{eq. 3})$$

These calculations define the number of rows required for effective separation and ensure the device can capture particles across the full range of interest. The row to row shift determines the distance that each subsequent row of posts should be

shifted laterally. The calculations also guided the choice of gap and post sizes (200 μm) to balance throughput and minimize clogging.

7.4 COMSOL Modeling

COMSOL Multiphysics was used to simulate the DLD device and verify particle separation. A simplified “Flow Past a Cylinder” model was adapted, with particles of 50 μm and 30 μm radius representing syncytial knots and smaller contaminants. The simulation tracked particle trajectories under the defined flow rate and channel geometry. Results showed successful lateral displacement of the larger particles while smaller particles followed the flow path, confirming the feasibility of the separation mechanism.

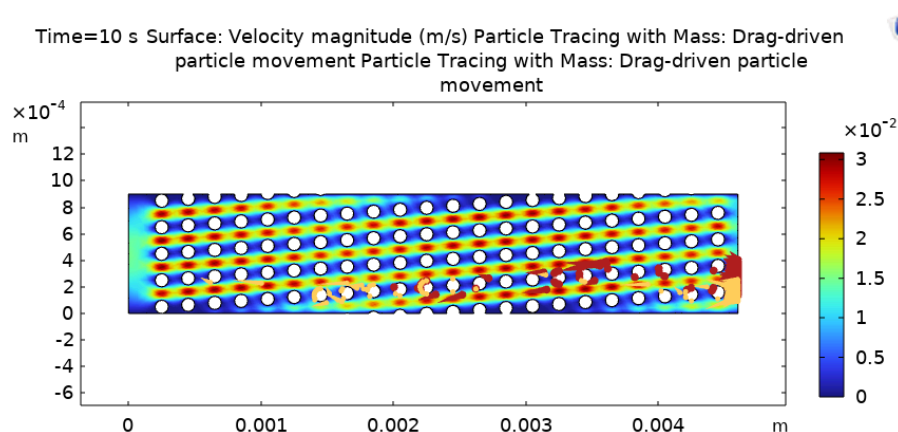


Figure 7. DLD COMSOL Modeling Results

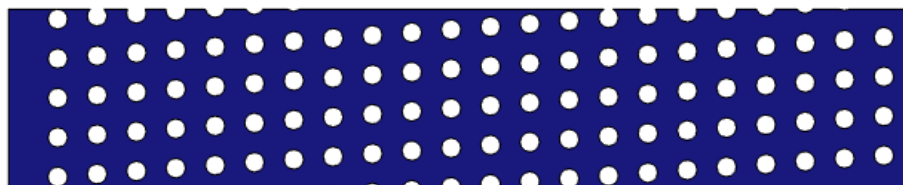


Figure 8. DLD COMSOL Modeling Animation

Figure 7 and 8 show that the red 50 μm and 30 μm particles are successfully separated in the microfluidic. The 50 μm particles representing syncytial knots are displaced along the y-axis and the 30 μm particles representing contaminants move laterally without being displaced. Based on these results we would like to scale up the microfluidic to encapsulate the full range of syncytial knots.

7.5 CAD Designs

CAD models of the DLD microfluidic were created based on the calculated parameters. The design includes multiple inlet channels designated for buffer and placental media and outlet channels to separate the syncytial knots from the small contaminants. These CAD designs serve as a template for fabrication. Figure 9 shows the arrangement of inlet and outlet channels. Inlet 1 and 3 will contain PBS as a buffer to maintain flow and inlet 2 will contain microspheres or placenta media. As particles flow laterally through the device the knots will get bumped towards the inlet further away, so outlet 4 is expected to contain the large particles representing syncytial knots and outlet 5 is expected to contain the small particles representing contaminants.

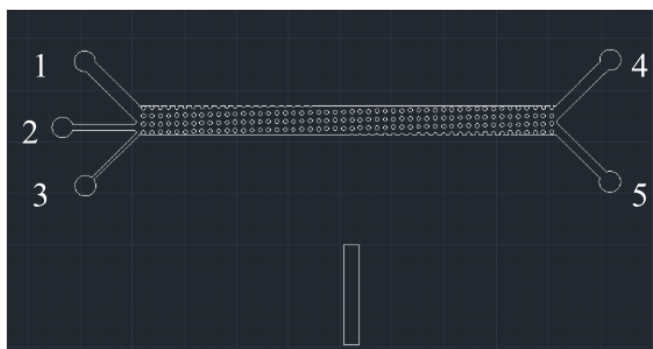


Figure 9. Detailed CAD view of DLD microfluidic with inlet and outlet channels labeled

Figure 10 shows the full design of all five DLD microfluidics. Design 1, 2, and 3 adjust for the number of rows in increasing order. Design 1, 4, and 5 adjust for the gap size in decreasing order.

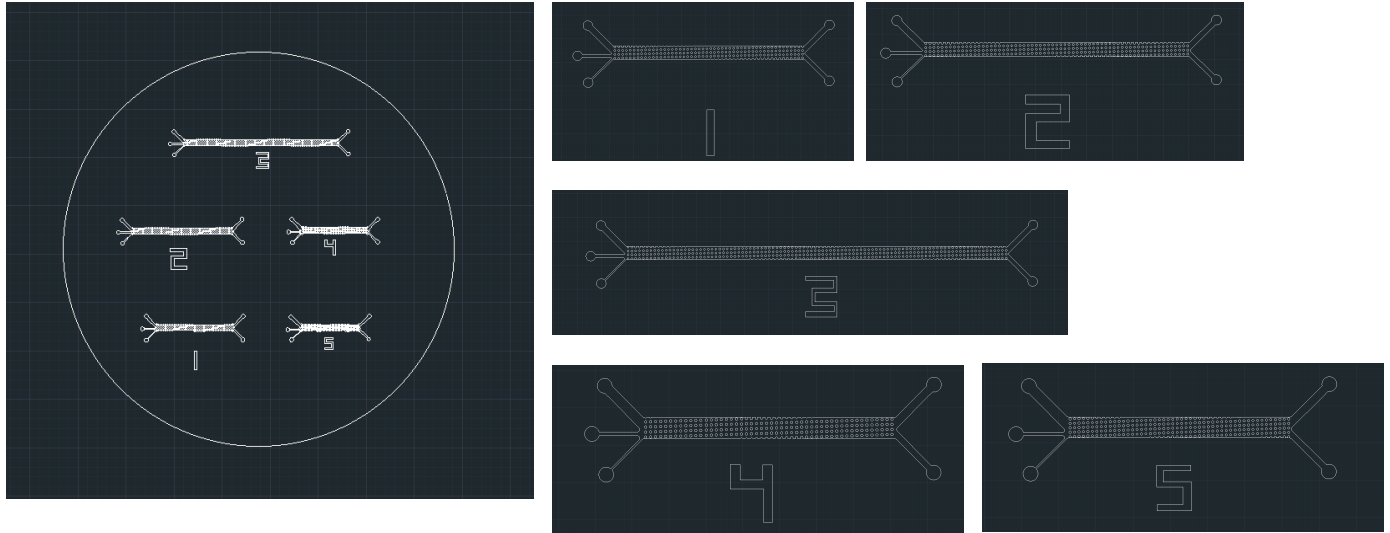


Figure 10. DLD CAD Designs for Experimentation

Figure 11 details the differences in the CAD designs for the DLD microfluidic.

The top comparison shows the differences in design when increasing the number of rows which lengthens the device. The bottom comparison shows the effect of reducing the gap size which minimizes the space between pillars.

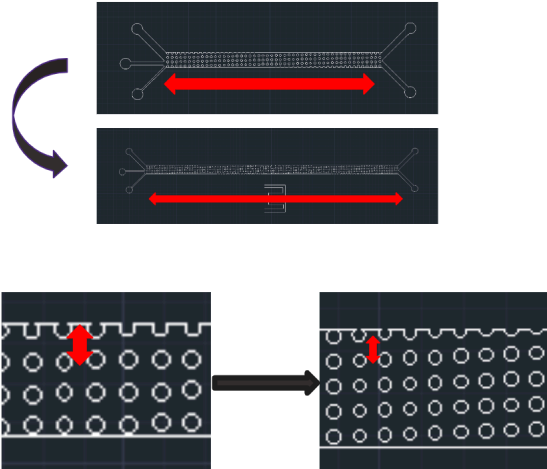


Figure 11. Comparison of DLD devices highlighting the differences between number of rows (top) and gap size (bottom)

The arrangement of rows and gap sizes included in the CAD designs are detailed in Table 3. By varying the number of rows and gap sizes with a low, medium, and high effect the impact of these parameters can be studied to find an optimal value.

Table 3. CAD Design Parameters

Design #	Gap Size	Number of Rows
1	200 μ m	50 rows
2	200 μ m	65 rows
3	200 μ m	100 rows
4	170 μ m	50 rows
5	150 μ m	50 rows

7.6 CAD Design Rationale

Design 1 was assigned a gap size of 200 μm to accommodate the full range of syncytial knot sizes (50–150 μm), providing additional clearance for the largest particles. The number of rows was calculated based on the distance required to laterally shift particles larger than the critical diameter by 300 μm , the minimum distance needed to direct them into separate outlet channels. This calculation indicated a minimum of 46 rows, which was rounded to 50.

To explore the effects of design parameters on separation efficiency, low, medium, and high levels were selected for both gap size and number of rows. The number of rows was set at 50, 65, and 100, with the middle value chosen closer to 50 to reflect the decreasing effect on separation efficiency expected as row count increases. Similarly, gap sizes were selected to test low (150 μm), medium (170 μm), and high (200 μm) values on separation efficiency. Gap size has a nonlinear effect on separation where values closer to the critical diameter enhance efficiency by reducing variability, so the middle value (170 μm) was chosen closer to the lower end to better capture this effect.

This combination of gap and row sizes allows systematic evaluation of the DLD design across the predicted operating range while efficiently exploring both low and high effect parameter spaces on separation efficiency.

8. MICROFLUIDIC FABRICATION PROCESS

The production of our DLD microfluidic device relies on standard soft lithography techniques, as illustrated in Figure 12. This approach allows for the high resolution replication of the pillar arrays necessary to achieve precise size based separation of syncytial knots.

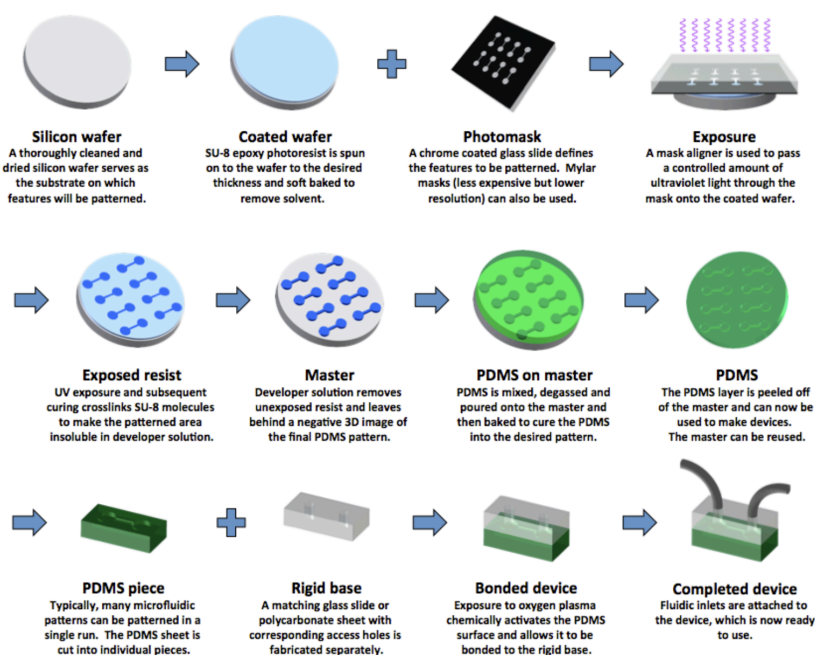


Figure 12. Microfluidic chip fabrication process³²

8.1 Mask Fabrication

The process begins with the creation of a master mold using photolithography. A silicon wafer is first cleaned and then spin-coated with SU-8, a light-sensitive photoresist³³. We carefully calibrated the spin speed to ensure the photoresist thickness matches our target channel height (200 μm). A high resolution transparency mask of our CAD designed DLD pillar patterns was aligned over the wafer and exposed to UV light. The UV light caused the polymer to harden only in the areas where the mask allowed the light to pass through, effectively locking

32. PDMS Microfluidics Fabrication | Nano- and Microscale Fabrication and Characterization. [accessed 2026 Mar 19]. <https://cleanroom.soe.ucsc.edu/microfluidics>

33. Microfluidic Pneumatic Cages: A Novel Approach for In-chip Crystal Trapping, Manipulation and Controlled Chemical Treatment. [accessed 2026 Mar 19]. <https://app.jove.com/t/54193/microfluidic-pneumatic-cages-novel-approach-for-chip-crystal-trapping>

our channel design onto the wafer. We then used a developer solution to wash away the soft, unexposed section, leaving behind a raised 3D mold of the pillar array. The finished master or mask acted as a permanent master template, allowing us to fabricate multiple identical chips from a single wafer.

8.2 PDMS Preparation, Pouring, Demolding, and Curing

The polydimethylsiloxane (PDMS) Silicone Elastomer base and the curing agent was mixed in a 10:1 ratio by weight³⁴. This specific ratio is critical for providing the chip with enough flexibility to be handled while remaining rigid enough to ensure stable flow. Following the mixing process, the mask is placed inside a petri dish and the PDMS added on top. The petri dish is then placed in a vacuum desiccator for degassing. This step is vital to remove any air bubbles that might obstruct the channels or interfere with the optical clarity needed for microscopy. The PDMS is then cured in an oven at 60°C for at least 2 hours. Once the material has fully polymerized and hardened, the PDMS slab is carefully demolded from the petri dish and separated from the mask, leaving a detailed negative impression of the DLD design.

8.3 Microfluidic Chip Assembly

The cured PDMS is cut and a biopsy punch is used to create the inlet and outlet ports. To seal the channels, the cut PDMS slab is cleaned using duct tape and a glass slide is cleaned using 70% ethanol and both are placed in an oxygen plasma cleaner³⁴. This chemical activation allows the two surfaces to form a permanent, covalent bond when pressed together, capable of sustaining the high flow rates required.

³⁴.PDMS Microfabrication and Design for Microfluidics and Sustainable Energy Application: Review. [accessed 2026 Mar 19]. <https://www.mdpi.com/2072-666X/12/11/1350>

9. DESIGN OF EXPERIMENTS

9.1 Experimental Goals

The overall goal of this study is to evaluate whether the DLD microfluidic devices can effectively separate syncytial knot sized particles from smaller contaminant particles. Using polystyrene microspheres as model particles, we aim to identify the optimal device design parameters, flow rates, and particle concentrations to inform future testing with placental tissue. Specifically, this quarter we focused on two design requirements: increasing throughput and automating the isolation process.

9.2 Model System

Polystyrene microspheres are used as a model for syncytial knots. Microspheres of defined sizes allow controlled, reproducible testing of device performance under varying flow rates and concentrations. Microspheres were diluted in PBS to the desired concentration with a fixed total volume depending on the experiment performed, facilitating direct comparison of the microsphere's pathway and separation efficiency across experiments. Table 4 details the corresponding particles being modeled by the microspheres.

Table 4. Microsphere sizes used to model syncytial knots and contaminants in experiments

Microsphere sizes (μm)	Corresponding Particle
50, 100	Syncytial knots
30, 10	Contaminants

9.3 Variables Tested

The key experimental variables are:

- **Device design:** Gap size and number of rows influence particle separation efficiency.
- **Flow rate:** Low, medium, and high rates evaluate the effect of fluid velocity on separation and throughput.
- **Microsphere concentration:** Low, medium, and high densities assess how particle density affects clogging and separation purity.

9.4 Experimental Setup

Each syringe pump was loaded with a fixed volume of solution (either the microsphere suspension or PBS buffer) and set to the target flow rate. Prior to operation, the microfluidic channels were prefilled with PBS to prevent air bubbles as shown in Figure 13. The syringes were then loaded onto the syringe pumps and viewed under a microscope.

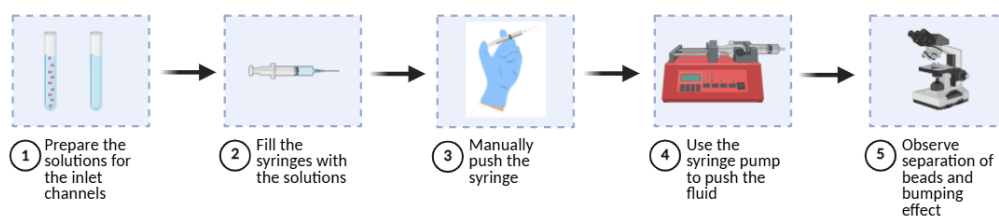


Figure 13. Experimental Setup

The top and bottom inlet contain PBS as a buffer and the middle inlet contains microspheres and in future testing will contain placenta media. Microsphere flow through the device was observed and recorded via microscopy, and particles were collected at the two outlet channels for analysis as shown in Figure 14.

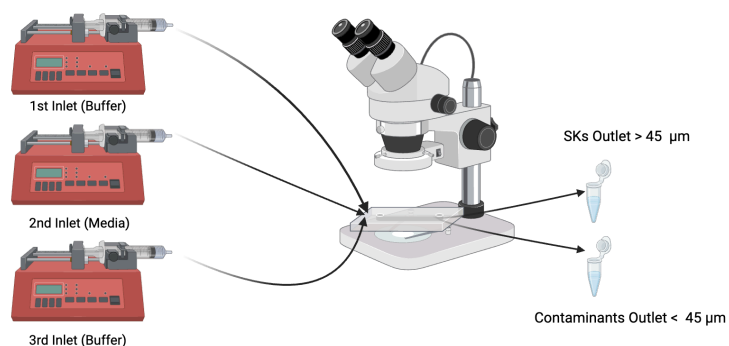


Figure 14. Syringe Pump Setup

9.5 Testing Plan

Each of the three prioritized designs (1, 2, and 4) would be tested in six trials as shown in Figure 15, resulting in 18 total experiments. The trials varied flow rate and microsphere concentration to assess their impact on transit time, clogging, and separation efficiency. Each parameter contained a low, medium, and high level to assess the impact on separation efficiency with a range of parameters. Trials 3 and 4 were repeated center points to assess experimental variability. The types of experiments were carefully selected to minimize the number of experiments to run and to account for confounding effects between the design elements, flow rate, and concentration.

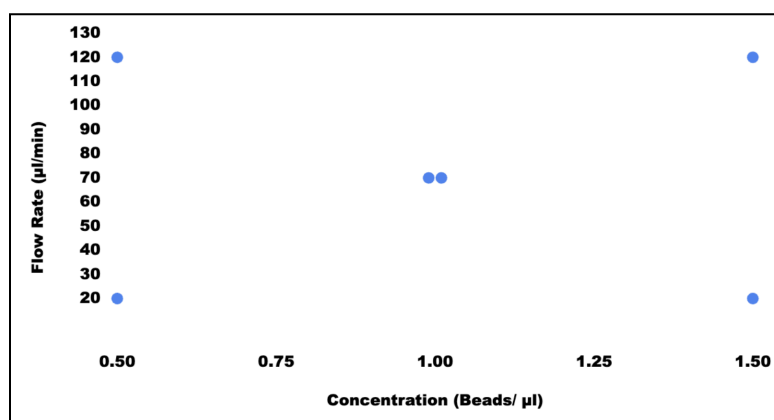


Figure 15. Overview of Testing Plan

9.6 Data Analysis Approach

Data analysis focuses on the following metrics:

1. **Theoretical vs. Observed Transit Rates:** Compare measured microsphere flow rates at the outlets with predicted values using microscopy.
2. **Clogging Assessment:** Qualitative comparison of the device throughout the experiments to assess for clumps of microspheres or reduced flow.
3. **Separation Purity:** Use ImageJ to quantify the percentage of target microspheres in the correct outlet and a hemocytometer to calculate relative purity.
4. **Processing Accuracy and Speed:** Evaluate effects of device geometry, flow rate, and microsphere concentration on throughput and purity.
5. **Best Device Selection:** One-way ANOVA followed by Tukey's HSD post hoc testing identifies the device with statistically superior performance in each category.

9.7 Success Metrics

Experiments are considered successful if:

- Particle flow rates are within $\pm 5\%$ of theoretical predictions.
- At least one device achieves $\geq 90\%$ separation purity.
- Minimal clogging occurs, indicated by outlet volumes near expected values and minimal accumulation in channels.
- Statistical analysis confirms significant differences in performance among devices.

9.7.1 Success Metrics Rationale

Multiple microfluidics and DLD specific studies have demonstrated that agreement within approximately $\pm 5\%$ between experimental and theoretical predictions is indicative of proper device operation³⁵. For example, DLD approaches that have achieved prediction errors of less than 5% in particle trajectories based on selected critical diameters reflect high reproducibility under controlled conditions³⁶. This is because DLD systems operate in a low Reynolds number regime, so flow behavior is highly predictable and small deviations can be attributed to fabrication tolerances and measurement uncertainty³⁷. However, larger deviations can significantly alter streamline structure and particle trajectories, leading to breakdown of deterministic separation behavior³⁸. Therefore, a $\pm 5\%$ tolerance is appropriate as it aligns with reported accuracy in DLD literature while remaining sufficiently strict to detect non-ideal effects such as clogging, channel deformation, or flow instability.

The theoretical prediction was determined by calculating the effective fluid volume of the chip, which was found by subtracting the total volume occupied by the pillar array from the overall channel volume. Next, the time it takes to fill the device was calculated based on the flow rate. This helped us find the exact amount of microspheres to pull from our stock to hit the target of 3 microspheres per second. We also accounted for 3.5x

35. Huang L. 2021. Microfluidic Flow Sensing Approaches. In: *Advances in Microfluidics and Nanofluids*: IntechOpen; [accessed 2026 Mar 19]. doi:10.5772/intechopen.96096

36. McGrath J, Jimenez M, Bridle H. 2014. Deterministic lateral displacement for particle separation: a review. *Lab on a Chip*. 14(21):4139–4158. doi:10.1039/C4LC00939H

37. Dincau B, Aghilinejad A, Kim J-H, Chen X. Characterizing the High Reynolds Number Regime for Deterministic Lateral Displacement (DLD) Devices. In: [place unknown]; [accessed 2026 Mar 19]. doi:10.1115/IMECE2017-71892

38. Zhang Z, Henry E, Gompper G, Fedosov DA. 2015. Behavior of rigid and deformable particles in deterministic lateral displacement devices with different post shapes. *J Chem Phys*. [accessed 2026 Mar 19] 143(24). doi:10.1063/1.4937171

dilution that happens when the sample mixes with the buffer inlets to keep the final count inside the chip accurate for verification. Setting the flow this way ensures the microspheres are spaced out enough to clearly see the displacement mechanism without the clumping or interference that happens at higher concentrations.

9.8 Contingency Plans

If successful criteria are not met, the following corrective actions (Table 5) will be taken to address the potential unexpected outcomes:

Table 5. Corrective methods for unexpected outcomes

Outcome	Corrective methods
Purity is <90%	Adjust flow rate and microsphere concentration; test other designs (3 and 5)
Clogging in all trials	Reduce flow rates and/or microsphere concentration
Theoretical vs. experimental discrepancy >5%	Particle concentration and flow rate will be systematically manipulated to evaluate whether a stable throughput of approximately 3 microspheres per second can be achieved.

10. RISK ANALYSIS

10.1 Chip Manufacturing

Chip manufacturing poses several potential failure modes, including incorrect PDMS base to curing agent ratios, incomplete degassing, and improper plasma bonding as shown in Table 6. These issues can result in difficult demolding, trapped air bubbles, collapsed channels, or blockages that reduce device functionality. Current controls include digital scales for accurate measurement, manual air bubble removal, and careful application of pressure during bonding. Corrective actions focus on careful preparation, degassing, and gentle handling to ensure channel integrity. The most severe failures here are channel collapse and blockages, which directly prevent particle flow and render a device unusable.

Table 6. Failure modes associated with chip manufacturing

Failure Mode	Effects of Failure	Mechanisms of Failure	Current Controls	S	O	D	RPN	Recommended Corrective Actions
Incorrect ratio of PDMS base to curing agent	Can make the removal process difficult	Human error during measurement	Digital scales for measurement	3	1	1	3	Ensuring the device is tared
Incomplete degassing	Air bubbles in the PDMS	Failure to remove small air bubbles	Manual removal of air bubbles with pipette tips	3	1	1	3	Keep the PDMS out for a few minutes before baking to ensure all air bubbles are removed.
Improper plasma bonding	Blockages of flow	Channels collapse	Manual hand pressure applied	4	1	1	4	Roll the PDMS gently onto the glass

10.2 Pump and Flow Control

Failures related to pump operation primarily include automatic withdrawal failure and inconsistent flow rates, which can lead to the introduction of air bubbles,

uneven fluid distribution, or insufficient throughput as shown in Table 7. Current controls include manual withdrawal of fluid to remove bubbles and checking flow rate settings. Corrective actions involve adjusting flow rates and adding extra PBS to push out trapped air.

Table 7. Failure modes associated with the pump

Failure Mode	Effects of Failure	Mechanisms of Failure	Current Controls	S	O	D	RPN	Recommended Corrective Actions
Automatic withdraw failure	Cannot withdraw automatically to reduce bubble formation	The pump is physically broken; cannot pull back properly	Manual withdraw of the sample into the syringe	3	3	1	9	Add extra PBS so more can be pushed out to ensure all air bubbles are removed

10.3 Syringe Preparation and Tubing

Syringe and tubing handling risks include microsphere settling, incorrect flow rates, insufficient fluid volume, and debris entering the tubing as shown in Table 8. These issues can lead to uneven particle distribution, clogging, or misrepresentation of device performance. Controls include using a shaker, volume and flow rate checks, and cleaning procedures. Corrective measures focus on manual microsphere withdrawal, potentially adjusting solution density, and thorough cleaning. The highest risks in this category are microsphere settling and debris clogging, which can significantly reduce separation efficiency and contaminate samples.

Table 8. Failure modes associated with syringe preparation and tubing

Failure Mode	Effects of Failure	Mechanisms of Failure	Current Controls	S	O	D	RPN	Recommended Corrective Actions
Settling of the microsphere	The microspheres	Gravity driven	Laboratory Shakers	4	2	2	16	Withdrawing the microspheres

	are not evenly distributed	sedimentation						manually instead of using the syringe pump to fill it up, change solution density
Incorrect flow rate	Uneven mixing of PBS and microspheres inside the device	Flow speed prevents the PBS and microspheres from mixing evenly	Checking the flow rate	4	1	4	16	Always double check the flow rate making sure that the correct flow rate corresponds to the right inlet
Too little fluid volume in a large syringe	Increased settling of particles	Increased distance particles travel to enter the microfluidic	Increasing the fluid volume	2	3	1	6	Increasing the fluid volume
Debris	Blocks fluid from entering the channel	Reduces the channel width which increases flow resistance in the channel thereby causing blockage	Minimal cleaning of the blade	5	1	2	10	Use a clean blade and store. Make sure to wipe any material that comes in contact with the setup with ethanol before proceeding
Settling of microspheres in the syringe and tubing contact point	Clogs the microfluidic	Sediment accumulates in the tubing and goes into the microfluidic	Laboratory Shakers	4	3	1	12	Use a combination of manual withdrawal with the laboratory shakers.

10.4 Inlet and Outlet Design

Inlet and outlet failures include improper tubing insertion, incomplete PDMS punctures, air bubbles near outlets, and insufficient rows for effective separation as shown in Table 9. These problems can prevent fluid from flowing correctly, reduce particle separation purity, or lead to inaccurate experimental results. Controls involve careful tubing placement, prefilling channels with PBS, and ensuring sufficient rows in the device. Corrective actions include adjusting row number, modifying gap size, flushing air bubbles, and repositioning tubing. The

most severe risks in this category are blocked inlets and bubbles near outlets, as they can stop flow entirely or drastically reduce syncytial knot separation efficiency.

Table 9. Failure modes associated with the inlet and outlets

Failure Mode	Effects of Failure	Mechanisms of Failure	Current Controls	S	O	D	RPN	Recommended Corrective Actions
Fluid is not flowing properly through the inlet channel	Fluid and particles get trapped at the inlet and are unable to move	Tubing is not fully inserted into the inlet channel	Use tweezers to push the tubing at least halfway into the channel	4	1	1	4	Make sure the PDMS is fully punctured before bonding and make sure the tubing is as far down the channel as possible
		PDMS is not fully punctured in the inlet channel	Make sure the PDMS is pushed out of the chip before bonding					
Particles >45 μm are not getting separated into the syncytial knot outlet	Separation of syncytial knots does not occur	Bumping is not occurring with the large particles	Having more rows than the minimum needed for separation	5	2	2	20	Change the gap size or number of rows to increase the bumping effect
Particles and contaminants are collected in the syncytial knot outlet	Purity of the collected sample is reduced	Flow patterns of the fluid allow small particles to get shifted along the device	Having buffers channels at the inlet to maintain a lateral streamline	3	2	2	12	Change the arrangement of inlet channels or reduce the flow rate
Most particles >45 μm are collected in the syncytial knot outlet, but some are in the contaminant outlet	The number of syncytial knots per placenta tissue collected is reduced	Too low number of rows	Having a larger than necessary number of rows for the desired shift	1	3	2	6	Increase the number of rows

Particles are not exiting at a certain outlet channel	Inaccurate results of particle endpoints	Air bubble near an outlet channel	Flush the microfluidic with PBS	5	2	1	10	Determine the cause of the air bubble and try to flush with PBS and make sure there is no air in the tubing, add a bubble trap to the inlet channel
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10.5 Severe Risk Analysis

A major source of variability in these experiments is the settling of microspheres within the syringe and tubing, which leads to inconsistent particle concentrations entering the device over time. This directly affects measurements of throughput and separation efficiency, as fluctuations in microsphere density can either overload the device or reduce particle interactions with the post array. Settling also increases the risk of clogging at narrow gaps which can halt flow entirely and invalidate experimental runs. Closely related to this is the risk of incorrect flow rates, which disrupt the precise fluid dynamics required for deterministic lateral displacement. Excessively high flow rates can prevent particles from properly interacting with the posts, while low flow rates can exacerbate settling and prolong processing time. Because flow rate influences all particles simultaneously, errors in this parameter can compromise the results and obscure the effects of key design variables.

These issues directly contribute to one of the most critical failures: particles not being separated into the syncytial knot outlet due to insufficient bumping. Effective separation depends on particles larger than the critical diameter being laterally displaced by the post array. If flow conditions or device geometry are not properly selected, these particles may follow unintended streamlines and mix with

contaminants. This reduces sample purity and undermines the goal for separation using the microfluidic devices, ultimately limiting the ability to achieve the target $\geq 90\%$ purity and desired throughput. Together, microsphere settling, flow rate inaccuracies, and ineffective bumping represent interconnected risks that can significantly impact device performance, highlighting the importance of careful experimental control and design optimization.

11. EXPERIMENTAL RESULTS & KEY FINDINGS

The following symbols (Table 10) are used to document the key observations throughout testing. These symbols allow consistent annotation of clogging, air bubbles, microsphere behavior, and bumping during experiments. A detailed visualization of the observations included in Table 10 are provided in Appendix A.

Table 10. Key for symbols correlating with testing observations

Observation	Symbol
Clogging	
Air bubbles	
No bubbles	
Microspheres in both outlets	
Microspheres not visible	
Bumping behavior observed	

Channel labeling conventions for inlets and outlets are shown in Figure 16, with inlets labeled 1-3 and outlets labeled 4-5.

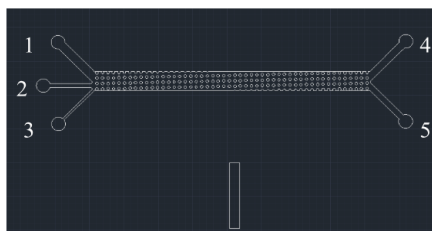


Figure 16. DLD microfluidic with inlet and outlet channels labeled

11.1 Initial Testing

Initial testing focused on validating DLD behavior using a three inlet system and polystyrene microspheres as shown in Table 11.

Guiding Question: Does the three inlet DLD configuration produce predictable bumping of particles above the critical diameter?

Summary of Outcome: Initial experiments using polystyrene microspheres revealed significant challenges, including air bubble formation, microsphere aggregation at high concentrations, and inconsistent flow. These factors prevented reliable observation of size based separation.

Design Decisions: The results indicated that the three inlet configuration introduced instability of flow due to the formation of air bubbles. Modifications to microsphere concentration, manual prefilling, and the addition of a bubble trap were proposed to mitigate these issues.

Table 11. Initial testing experimental results

Experiment	Setup	Observations	Interpretation	Design Decision
100 μ m microspheres and PBS	-Inlet 1 and 3: PBS -Inlet 2: 1mL microspheres at 1 microsphere/ μ L	 -Microspheres that were heading towards outlet 5 moved through the gap to outlet 4	-Concentration of microspheres too high and particle interactions led to clumping -Flow could not be accurately modeled due to air bubbles	Reduce the concentration of microspheres, use a vibrational shaker, order a bubble trap to remove air bubbles from inlet 2
30 μ m microspheres and PBS	-Inlet 1 and 3: PBS -Inlet 2: 1mL microspheres at 1.5 microsphere/ μ L	 -Some of the microspheres got stuck to the pillars	-Running the microspheres inlet incorporated air bubbles -Particle interactions with the pillars	Test the flow pattern with dye to see if there is a pattern to air bubble formation

Dyed PBS and 3 Inlet Channels	-Inlet 1 and 3: red -Inlet 2: green		-Air bubble formation is influenced by 3 inlet channels	Test 2 inlets at a time and figure out the best configuration for running the device to minimize air bubbles
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11.2 Effect of Inlet Design

To address the limitations observed in initial testing, the effect of inlet configuration on flow stability was systematically evaluated as shown in Table 12.

Guiding Question: Which inlet configuration produces the most stable flow with minimal air bubbles?

Summary of Outcome: Reducing the number of active inlets improved flow stability, with the 1 & 3 inlet configuration producing the fewest bubbles.

Design Decisions: Two inlet systems were adopted for subsequent experiments to reduce bubble formation and improve reproducibility.

Table 12. Testing the effect of inlet design results

Experiment	Setup	Observations
Inlets 1 and 2	-Chip #2 and #4 -Inlet 1: yellow -Inlet 2: red -All inlets: 70 μ L/min	
Inlets 2 and 3	-Chip #1 -Inlet 2: yellow -Inlet 3: red -All inlets: 70 μ L/min	 -Changing the flow rate didn't affect the air bubbles (35 and 140 μ L/min)
Inlets 1 and 3	-Chip #3 and #5 -Inlet 2: yellow -Inlet 3: red -All inlets: 70 μ L/min	 -Changing the flow rate didn't affect the air bubbles (35 and 140 μ L/min) -Had the fewest air bubbles overall

11.3 Microsphere Testing with Two Inlet System and Manual Pushing

Following the identification of a more stable inlet configuration, microsphere experiments were conducted using a modified two inlet system as shown in Table 13.

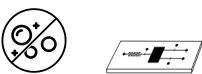
Guiding Question: Does the two inlet configuration improve particle separation and bumping behavior?

Summary of Outcome: The two inlet system did not improve flow, but the addition of manually pushing the syringe increased flow stability. Uneven distribution of microspheres was observed, with most particles exiting a single outlet with the manual pushing.

Design Decisions: Further optimization of operating parameters (flow rate, microsphere concentration) and potential manual flushing are necessary to remove air bubbles in order to achieve consistent bumping behavior.

Table 13. Testing with microspheres and manual flushing results

Experiment	Setup	Observations
Dye	Chip #1	Very little occurrence of air bubbles
100 μm	Chip #2,4 and 5	
50 μm	Chip #2,4 and 5	 -Syncytial knot outlet had majority of the liquid while the contaminants outlet had very little
100 μm	Manual syringe	-Reduced air bubbles -Microspheres settling due to size -Debris
50 μm	Manual syringe	 

30 μm	Manual syringe	
50 μm + 30 μm + 10 μm	Manual syringe	 -Most of the microspheres of all sizes were exiting from the first outlet (SK)

11.4 Flow Control with Manual and Syringe Pump Combination

To further improve flow stability and reduce air bubble formation, a combination of manual loading and syringe pump control was implemented. We ran the test plan, but some trials were impacted by human error related to how the tubing was placed and sealed at the inlets and outlets. In these instances, flow was only coming out of one outlet instead of both, which prevented us from getting a full set of results for the data table.

Guiding Question: Can combining manual loading and syringe pumps further stabilize flow and reduce bubbles while maintaining an expected flow pattern to drive bumping behavior?

Summary of Outcome: Manual prefilling reduced trapped air and improved flow stability compared to syringe only operation. Microsphere settling and debris introduction remained challenges. Out of the six testing conditions 1.5 microspheres/ μL concentration and 120 $\mu\text{L}/\text{min}$ flow rate were the most promising because there was a steady flow of the microspheres in the device and clear bumping was observed. In the future, more runs of the test plan are necessary to account for this variability and to determine which parameters are best for the final verification of the device

Design Decisions: Sample preparation will continue to include manual prefilling the device with PBS and then using the syringe pump to maintain the desired flow rate. Additional measures of air bubble control will be incorporated including a bubble trap.

11.5 Additional Experimental Data

Detailed information regarding the results of the experiments and visual evidence of the failure modes are detailed in Appendix A: Additional Data for the Experimental Results.

12. DISCUSSION

12.1 Summary of Results vs Metrics

The experimental results demonstrate partial progress toward the design requirements outlined in Section 3.2, particularly in improving flow stability and increasing automation. While successful separation at the target $\geq 90\%$ purity was not consistently achieved, the experiments identified key factors affecting performance, including inlet configuration, flow rate, and particle concentration. Improvements such as reducing inlet number and prefilling the device helped minimize air bubbles, contributing to more stable flow conditions. However, throughput and separation efficiency remained below target levels due to inconsistent bumping behavior and uneven particle distribution across outlets.

12.2 Design Tradeoffs

Several design tradeoffs emerged during testing, particularly between flow stability, separation efficiency, and system complexity. Increasing flow rates helped reduce air bubble formation but risked compromising separation by increasing pressure so that small particles were also displaced. Similarly, reducing the number of inlets improved flow stability but introduced challenges in maintaining balanced pressure and equal outlet distribution. These tradeoffs highlight the need to balance operational stability with precise control of particle trajectories.

12.3 Limitations

This study has several limitations that impact the interpretation of results. First, the use of polystyrene microspheres, while useful for controlled testing, does not

fully replicate the biological variability and deformability of syncytial knots. Second, persistent issues such as air bubbles, microsphere settling, and debris introduced confounding variables that limited the ability to isolate the effects of individual design parameters. Additionally, the number of trials and device configurations tested was constrained by time and fabrication resources, limiting the extent of optimization. These factors collectively reduced the ability to achieve consistent and reproducible separation outcomes.

12.4 Key Insights

Despite these limitations, several important insights were gained. The inlet configuration was identified as a critical factor in maintaining stable flow, with the two inlet system providing an alternative to the initial three inlet design to reduce air bubbles. Prefilling the device and incorporating manual control steps significantly reduced air bubble formation, emphasizing the importance of preparation in microfluidic experiments. Additionally, the results highlighted that achieving effective bumping behavior requires careful coordination of flow rate, particle concentration, and device geometry. These insights will directly inform future design iterations, including the integration of bubble traps, optimization of gap size and row number, and refinement of operating conditions for testing with biological samples.

13. FUTURE WORK

13.1 Completion of DOE and Design Validation

Although a design of experiments (DOE) was developed to evaluate the effects of gap size, number of rows, flow rate, and particle concentration on separation efficiency, these tests were not completed due to flow instability and experimental challenges. As a result, key questions regarding the optimal gap size and number of rows remain unresolved.

While waiting for fabrication of the redesigned device, additional testing on the current designs with the parameters from the testing plan will serve to establish baseline trends. This will allow us to determine which concentrations and flow rates are most effective for the next stage of testing with the redesigned devices.

13.2 CAD Improvements and Modified DOE

Based on the experimental results where microspheres were exiting into unexpected outlet channels, several modifications to the CAD design are proposed to improve flow stability and outlet accuracy detailed in Figure 17.

Decreasing the angle at the outlet junction is expected to promote smoother streamline separation and reduce unintended particle crossover between channels.

Additionally, incorporating a physical barrier near the outlet region may prevent larger particles from entering the incorrect outlet, improving separation purity.

Newly designed devices will contain both 3 inlets and 2 inlets for a comparison of inlet effects since air bubbles formed with both combinations of inlets in the initial testing.

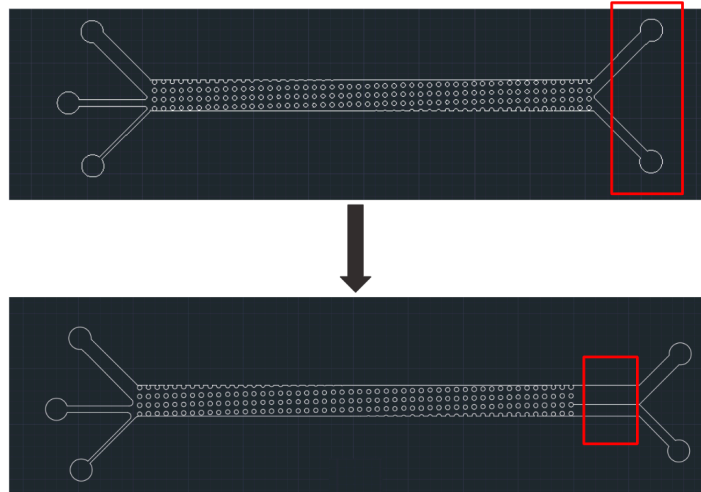


Figure 17. CAD comparison of the old DLD design (top) and the new proposed DLD design (bottom) that includes a straight channel 3mm long with branched outlet channels at 45°

These design changes alter flow behavior near the outlet and may impact overall device performance, requiring an updated DOE. The modified DOE will retain key variables such as flow rate and particle concentration, while refining geometric parameters to reflect the redesigned outlet region. Testing will focus more on the effect of gap size and number of rows rather than flow rate and concentration.

This new DOE approach will allow for more targeted identification of the configuration that provides optimal separation efficiency, minimal clogging, and stable flow conditions.

13.3 Bubble Trap Integration

Air bubble formation was identified as a major limitation affecting flow stability and device performance. While manual prefilling reduced bubble introduction, bubbles remained a concern, as those microbubbles that were already in the solution were carried into the syringe, propagated into the tubing and subsequently entered the microfluidic chip.

To address this, future experiments will integrate a bubble trap into the middle inlet to remove trapped air before it enters the microfluidic channels. With improved flow stability, experiments can be conducted under controlled and reproducible conditions, enabling completion of the DOE and accurate measurement of device performance.

13.4 Validation with Placental Explant Tissue

Following completion of the DOE, optimization of device geometry, and quantitative validation using microsphere models, the final phase of this project will involve testing with placental explant tissue containing syncytial knots.

Unlike microspheres, biological samples introduce additional complexities, including variability in particle size, deformability, and potential interactions with channel surfaces. Therefore, parameters identified during microsphere based testing such as optimal flow rate and concentration, will be adjusted to account for these differences.

This stage will serve as the final validation step to confirm that the device can reliably isolate intact syncytial knots while maintaining their structural integrity. Successful tissue testing will demonstrate the device's applicability for placental research and support its potential as a high throughput alternative to manual isolation methods.

14. CONCLUSION

Placental health is a critical indicator of maternal and fetal well being, yet current methods for isolating syncytial knots are limited by low efficiency, lack of automation, and difficulty separating them from similar sized cellular debris. This project addressed this challenge by developing and evaluating a deterministic lateral displacement (DLD) microfluidic device designed to isolate syncytial knot sized particles. By leveraging differences in particle size and flow behavior, the proposed system aims to provide a more controlled, scalable, and automated method for syncytial knot isolation compared to existing techniques.

Through iterative testing with polystyrene microspheres, this study established an initial proof of concept for the device while identifying key factors influencing performance. Results demonstrated that inlet configuration, flow rate, and particle concentration significantly impact flow stability and separation behavior. Transitioning from a three inlet to a two inlet system and incorporating manual prefilling improved flow consistency by reducing air bubbles. Challenges such as microsphere settling, inconsistent bumping behavior, and uneven outlet distribution limited the ability to achieve the target separation purity and throughput. Despite these limitations, the experiments provided critical insights into optimizing device geometry and operating conditions.

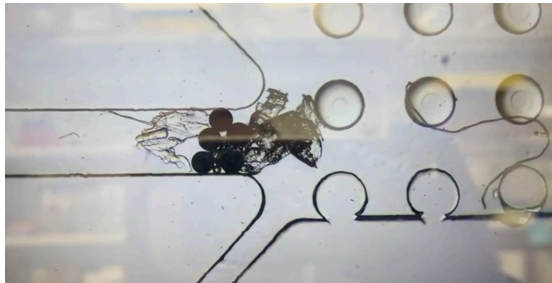
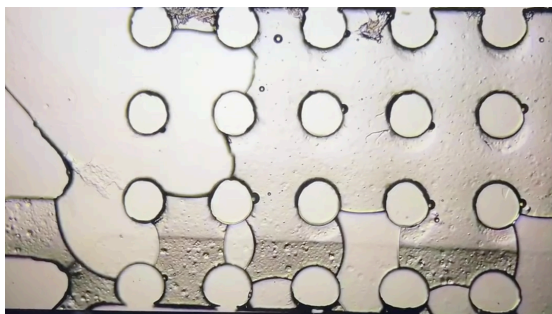
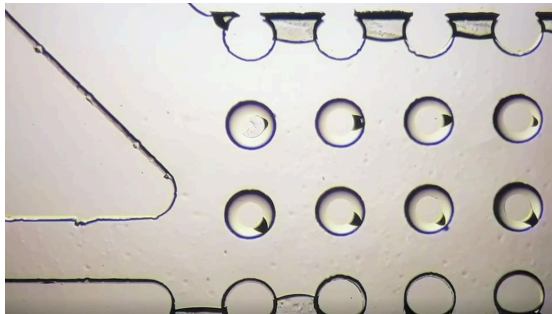
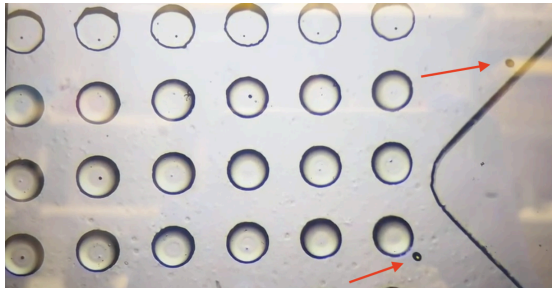
Overall, this work advances the development of a microfluidic platform for syncytial knot isolation by identifying key design constraints and validating core aspects of DLD behavior. While further optimization is required, including CAD redesign, bubble trap

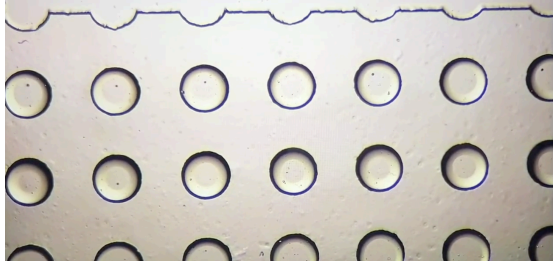
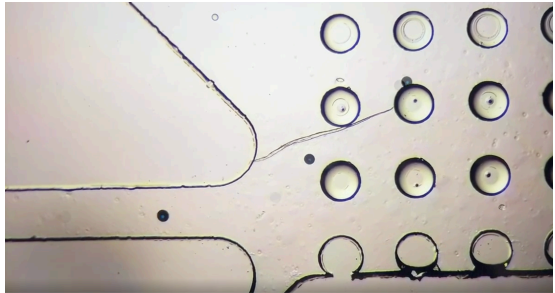
integration, and testing with biological samples, the results demonstrate a promising pathway toward a more efficient and automated approach to studying maternal and fetal health.

Appendix A: Additional Data for the Experimental Results

Appendix A

Table 14. Visualization of testing observations based on experimental results

Observation	Symbol	Image from experiment
Clogging		
Air bubbles		
No bubbles		
Microspheres in both outlets		

Microspheres not visible		
Bumping behavior observed		

1. Initial Testing

Initial device verification revealed challenges with air bubbles. The dye experiment visualizes where the bubbles form in the following image

 Air Bubbles (Initial Testing/ Dye Experiment) .png . Subsequent testing showed significant clogging within the device  Clogging.png . This was due to debris (potentially from human error or microsphere stock) and the 100 μm microspheres settling too quickly due to their size. To resolve this, a vibrational shaker was used to keep the microspheres in suspension. This final image shows the 100 μm microspheres remained suspended and did not settle in the syringe

 100 Microns w/ Vibrational Shaker.png .

2. Effect of Inlet Design

To minimize air bubbles, different combinations of inlets were tested. This video captures the best performing combination using inlets 1 and 3. While some air bubbles were still observed, this experiment produced the fewest air bubbles  Inlets 1 & 3.mov .

3. Microsphere Testing with Two Inlet System

In this video, a 50 μm microspheres is captured at a flow rate of 10 $\mu\text{L}/\text{min}$ moving through the array and successfully demonstrating bumping properties. However, it ultimately exits into the waste outlet instead of the SKs outlets  2 inlets.mov . This confirms the necessity of the redesigned CAD model to fix the outlet positioning.

- The failure to reach the correct outlet was also observed during the 3 inlet manual and syringe pump experiments  Incorrect Outlet (Manual Pump ONLY).mov .

4. Flow Control with Manual and Syringe Pump Combination

Before loading the microspheres, the device is prepared using a precise flushing strategy to ensure a bubble-free environment. Initially, the largest buffer inlet is run manually to clear the channels, followed by the attachment of all three inlets to the syringe pumps for synchronized flow.

- This video demonstrates the flow disruption and channel blockages that occur without a proper initial flush.  Initial Flush (Air Bubbles).mov
- This video captures the current preparation strategy, showing the manual pressure applied to the primary buffer inlet to successfully clear the device before the syringes are placed on the syringe pumps.  Initial Flush (Manual Flow).mov
- Following the flush, this video captures a specific experiment from the test plan at the highest concentration and flow rate using 50 μm microspheres. Consistent bumping without clogging is observed.  50 Microns.mov